New Generalization Bounds for Quantum Learning via f-Divergences and Maximal Leakage

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***Abstract*—We study high-probability generalization bounds for learning from classical-quantum data, via the framework of Caro et al. (COLT, 2024) and Warsi et al. (IEEE Trans. Inf. Theory, 2026), which separates the analyses into two layers: an inner quantum reference-tail bound, which accounts for quantum-state dependence in the loss, and an outer classical change-of-measure step, which accounts for the dependence between the training sample and the hypothesis. We refine both layers. For the classical layer, we use a data processing trick that is simple yet powerful, yielding bounds based on f-divergences and Sibson's information. For the quantum layer, we use the quantum hockey-stick divergence to sharpen the reference tail. We evaluate both cases via explicit examples, including a Bell-augmented QNN selection on real data and a multi-tier randomized-reveal channel. Both examples validate our improvements over existing bounds.**

***Index Terms*—Quantum machine learning, generalization error, f-divergence, maximal leakage, quantum information theory.**

A full version of this paper with proofs is accessible at [1].

I. Introduction

A randomized learning algorithm $P_{W|S}$ can be characterized by a probabilistic mapping from input S to output hypothesis W. A learning algorithm *generalizes* when its performance on a training sample persists on fresh data from the same population, as quantified by the *generalization error*. Low generalization error implies better generalization performance.

In the past decade, information-theoretic generalization bounds have been extensively studied [2]–[10], which usually quantify the dependence between the training sample and the output hypothesis via different information measures. There exist two major classes of generalization bounds, namely expectation bounds, which guarantee the algorithm's behavior averaged over the sample and algorithmic randomness, and single-draw bounds, which guarantee the tail probability of the generalization error for a single drawn hypothesis. We focus on the latter, for which Esposito et al. [5] utilized Rényi divergences, f-divergences, and maximal leakage, while [9] utilized change-of-measure inequalities [11]–[13] as a unified framework; also see [10] for differentially private algorithms.

More recently, information-theoretic generalization error bounds for quantum learning algorithms have been proposed [14]–[17]. Learning from quantum data requires a richer description because quantum examples are processed through quantum operations and accessed through measurements [14], [15], [18]. In general, quantum learning includes both quantum models for classical data [19], [20] and learning from quantum data, such as quantum-state learning [21] and learning from quantum example oracles [22]. We focus on the latter case.

By [14], [15], a classical-quantum learning framework with generalization analysis has been studied, in which a classical sample indexes quantum data with training and testing subsystems, and the learner can produce classical and quantum hypotheses. [15] proposed a true loss based on an independent testing sample and a conditionally averaged quantum hypothesis, and derived bounds using quantum Rényi divergences and smooth max-information. Also see [17] for a global classical-quantum information measure via sandwiched quantum Rényi divergence and [16] for quantum PAC-Bayesian bounds.

In this paper, we utilize a quantum learning framework of [15] for generalization bounds, in which a classical sample S indexes a quantum data state with training and testing subsystems, the learner performs a sample-dependent POVM on the training subsystem to produce a classical output W, and then applies a quantum channel to generate a quantum hypothesis state. This training-testing bipartition separates the learning operation from loss evaluation without requiring the two subsystems to be independent. The generalization analyses are divided into two parts: an outer classical layer and an inner quantum layer. We refine both layers, and we note that the latter relies on the quantum hockey-stick divergence [23], for which related work can be found in [24]–[27]. We use two concrete examples to quantify the effective improvements from the two refinements: a Bell-augmented QNN with stochastic model selection on a real dataset and a multi-tier channel in which we vary the rare-class mass.

Organization. Section II introduces the quantum learning framework. Section III provides refined generalization bounds for the classical layer, and the improvements are evaluated via two examples in Section IV. Section V then provides refinements for the quantum layer, for which the improvements are evaluated again in Section VI by revisiting the two examples.

II. Quantum Learning Model

We introduce the classical-quantum learning model considered in [14], [15], as illustrated in Fig. 1, where a quantum learning algorithm processes quantum data indexed by classical training samples and produces classical and quantum hypotheses. We later refine the generalization bounds of [15] through two layers, one is classical and another is quantum.

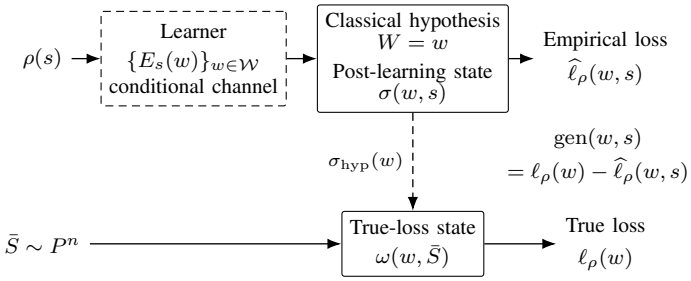


Fig. 1. The quantum learning model. The learner acts on the training subsystem of $\rho(s)$ and produces a classical hypothesis $W = w$ and conditional state $\sigma(w, s)$. The hypothesis marginal is $\sigma_{\rm hyp}(w) = \mathbf{E}_{S|W=w} \mathrm{Tr}_{\rm te}\, \sigma(w, S)$. For independent $\bar{S} \sim P^n$, the true-loss state is $\omega(w, \bar{s}) := \rho_{\rm te}(\bar{s}) \otimes \sigma_{\rm hyp}(w)$.

A. From Classical to Quantum Learning

In classical supervised learning [2], [3], let $S = (Z_1, \ldots, Z_n) \sim P^n$ be the training sample, where $Z_i \overset{\text{i.i.d.}}{\sim} P$. A randomized learning algorithm produces a classical *hypothesis* $W \in \mathcal{W}$ according to the induced distribution $P_{W|S}$. For a loss function $\ell : \mathcal{W} \times \mathcal{Z} \to \mathbb{R}$, the empirical loss and population loss are defined by $\widehat{\ell}_s(w) := \frac{1}{n}\sum_{i=1}^n \ell(w, z_i)$ and $\ell(w) := \mathbf{E}_{Z\sim P}\ell(w, Z)$, respectively. Fixing w, the *generalization error* is defined by $\ell(w) - \widehat{\ell}_s(w)$. Information-theoretic generalization bounds control the error by quantifying the dependence between S and W [2], [3], [5], [9].

In quantum learning, the learning algorithm retains a quantum hypothesis that depends on the training sample and remains correlated with the quantum systems used to evaluate the loss. Our framework accommodates both classical and quantum hypotheses and captures the resulting classical and quantum dependencies relevant to generalization [14], [15]: conditioned on $S = s$, the learner receives a quantum data state $\rho(s)$ on the test-training bipartition $\mathcal{H}_{\rm te} \otimes \mathcal{H}_{\rm tr}$. The two subsystems may be correlated or entangled. We write $\rho_{\rm tr}(s) := \mathrm{Tr}_{\rm te}\, \rho(s)$ and $\rho_{\rm te}(s) := \mathrm{Tr}_{\rm tr}\, \rho(s)$ for their marginals.

A quantum learner acts on the training subsystem and gives a classical output $W \in \mathcal{W}$ by a possibly sample-dependent POVM $\{E_s(w)\}_{w\in\mathcal{W}}$, and induces a channel $P_{W|S}(w|s) := \mathrm{Tr}\,[E_s(w)\rho_{\rm tr}(s)]$. Based on w, the learner may apply a conditional quantum channel to the training subsystem to obtain a quantum hypothesis. $\sigma(w, s)$ is the resulting joint state of the testing and hypothesis systems.[1] The hypothesis marginal is $\sigma_{\rm hyp}(w, s) := \mathrm{Tr}_{\rm te}\, \sigma(w, s)$. Fixing w does not in general make the retained quantum hypothesis independent of s.

B. Losses and Generalization

Similar to [15], we assume the associated quantum and classical sub-Gaussianity conditions with parameters μ and τ, respectively. Let $\widehat{L}(w, s)$ be a Hermitian loss observable on the post-learning test-hypothesis system. Its expectation under the actual post-learning state defines the *empirical loss*

$$\widehat{\ell}_\rho(w, s) := \mathrm{Tr}\left[\widehat{L}(w, s)\sigma(w, s)\right]. \tag{1}$$

[1] As in [15], we fix a measurable extension to pairs with $P_{W|S}(w|s) = 0$, so that fixed-w reference expectations under P^n are well defined.

We adopt the true loss in [15]. Let $\bar{S} \sim P^n$ be a fresh sample independent of the training sample and learning output. For every $w \in \mathrm{supp}(P_W)$, define the average hypothesis state $\sigma_{\rm hyp}(w) := \mathbf{E}_{S\sim P_{S|W=w}}[\sigma_{\rm hyp}(w, S)]$ and the *true loss* be

$$\ell_\rho(w) := \mathbf{E}_{\bar{S}\sim P^n} \mathrm{Tr}\left[\widehat{L}(w, \bar{S})\left(\rho_{\rm te}(\bar{S}) \otimes \sigma_{\rm hyp}(w)\right)\right], \tag{2}$$

where conditional average is via the given learning algorithm. Compared to [14], (2) is averaged over the training samples given $W = w$, rather than evaluated at the same fresh sample as the testing state. Thus, the testing sample and the quantum hypothesis in the true-loss product state are decoupled.

We denote $\omega(w, s) := \rho_{\rm te}(s) \otimes \sigma_{\rm hyp}(w)$; (1) and (2) compare the expectation of the loss observable under $\sigma(w, s)$ and $\omega(w, s)$, respectively. Define the *single-draw generalization* by

$$\mathrm{gen}(w, s) := \ell_\rho(w) - \widehat{\ell}_\rho(w, s). \tag{3}$$

In this paper, we focus on the *high-probability generalization bound* [5], [9], [15], [28], i.e., we seek a *radius* $r(\delta)$ satisfying

$$\Pr\{|\mathrm{gen}(W, S)| > r(\delta)\} \le \delta.$$

C. Reference Tails

Similar to [15], we separate the generalization error analysis into two layers. The *inner quantum layer* bounds the error for each fixed hypothesis w under P^n, and accounts for the quantum-state dependencies in the loss and the concentration of the empirical loss. The *outer classical layer* then transfers the bound to the actual law P_{WS}, accounting for the dependence between the training sample S and the hypothesis W.

Define $E_\epsilon := \{(w, s) : |\,\mathrm{gen}(w, s)| > \epsilon\}$ and $E_{\epsilon,w} := \{s : |\,\mathrm{gen}(w, s)| > \epsilon\}$. In the inner layer, a *quantum correction* is needed since the empirical and true losses are evaluated under different quantum states. Define $m_w := \mathbf{E}_{S\sim P^n}[\widehat{\ell}_\rho(w, S)]$ and $b_w := \ell_\rho(w) - m_w$. By (1) and (2), we have

$$\begin{aligned} b_w &= \mathbf{E}_{S\sim P^n} \mathrm{Tr}\left[\widehat{L}(w, S)\big(\omega(w, S) - \sigma(w, S)\big)\right], \\ \mathrm{gen}(w, S) &= b_w + m_w - \widehat{\ell}_\rho(w, S). \end{aligned} \tag{4}$$

Here, b_w captures quantum correlations between the testing and hypothesis systems, as well as the possible dependence of the quantum hypothesis on S even for fixed w. Bounding $|b_w|$ yields the quantum correction. The decomposition (4) thus suggests two refinements: a tighter quantum correction and a sharper concentration bound for the fixed-w empirical loss.

We use the reference-tail estimate of [15, Theorem 4]. For $\alpha \in (0, 1)$, let $c_1(\alpha, w)$ denote the quantum Rényi-divergence bound on $|b_w|$ derived therein. We consider the correction term $c^\star := \inf_{\alpha\in(0,1)} \sup_{w\in\mathrm{supp}(P_W)} c_1(\alpha, w)$ and have

$$\sup_{w\in\mathrm{supp}(P_W)} P^n(E_{\epsilon,w}) \le \theta_\epsilon := \min\left\{1, 2\exp\left(-\frac{n[(\epsilon - c^\star)_+]^2}{2\tau^2}\right)\right\}. \tag{5}$$

For an auxiliary law R_W such that $R_W \ll P_W$, define $Q_R := R_W P^n$, under which S, W are independent. By (5),

$$Q_R(E_\epsilon) = \mathbf{E}_{W\sim R_W} P^n(E_{\epsilon,W}) \le \theta_\epsilon. \tag{6}$$

The goal is to bound $P_{WS}(E_\epsilon)$ by δ for the smallest ϵ. We first refine the outer classical change-of-measure step from Q_R to P_{WS} in Section III. We then improve the inner reference-tail estimate using quantum hockey-stick divergence in Section V.

III. Classical Change-of-Measure Refinements

We first refine the classical layer (retaining the same $c^\star$) via a simple-yet-powerful data processing inequality (DPI) trick that is recently revisited for generalization bounds [9], [13].

Fix $\epsilon > 0$ and auxiliary R_W; let $Q_R := R_W P^n$. Define $p := P_{WS}(E_\epsilon)$ and $q_R := Q_R(E_\epsilon) = \mathbf{E}_{W\sim R_W} P^n(E_{\epsilon,W})$. By (5), we have $q_R \le \theta_\epsilon$. More generally, if $P^n(E_{\epsilon,w}) \le \vartheta_w(\epsilon)$, then $q_R \le \Theta_R(\epsilon) := \mathbf{E}_{W\sim R_W}[\vartheta_W(\epsilon)]$. This hypothesis-dependent form will be used in our quantum refinements in Section V.

Warsi et al. [15] take $R_W = P_W$ and use Rényi divergence and Hölder's inequality to derive $p \le e^{\frac{\beta-1}{\beta} J_\beta} q_R^{\frac{\beta-1}{\beta}}$ where $J_\beta := D_\beta(P_{WS}\|P^n P_W)$. Here we derive new bounds in terms of f-divergences and Sibson's information.

A. Generalization Bounds via f-Divergences

Let $f : (0,\infty) \to \mathbb{R}$ be strictly convex and continuous, $f(1) = 0$, and D_f denotes the f-divergence. For $u, v \in [0,1]$, define $d_f(u\|v) := D_f\big(\mathrm{Ber}(u)\|\,\mathrm{Ber}(v)\big)$. For $\theta \in [0,1]$ and $J \ge 0$, define $\mathfrak{R}_f(\theta, J) := \sup\{u \in [\theta,1] : d_f(u\|\theta) \le J\}$.

Rather than using Hölder's inequality [15], motivated by [9], we apply DPI to the indicator event $(w,s) \mapsto \mathbf{1}\{(w,s) \in E_\epsilon\}$ and obtain $D_f(P_{WS}\|Q_R) \ge d_f(p\|q_R)$. Together with $q_R \le \theta_\epsilon$, this yields an upper bound on p. This trick gives retains the probabilities of both E_ϵ and its complement and gives:

Theorem 1. *For finite $\mathcal{W}$ and R_W s.t. $\mathrm{supp}(R_W) = \mathrm{supp}(P_W)$, let $Q_R := R_W P^n, J_f(R_W) := D_f(P_{WS}\|Q_R)$; for every $\epsilon > 0$,*

$$P_{WS}(E_\epsilon) \le \mathfrak{R}_f\big(\theta_\epsilon, J_f(R_W)\big), \tag{7}$$

under the reference-tail estimate (6)*. If $d_f(\delta\|q) = J_f(R_W)$ has a solution $q_f^\star(\delta, J_f(R_W)) \in (0,\delta]$ for $\delta \in (0,1)$, then*

$$\Pr\Big\{|\,\mathrm{gen}(W,S)| > c^\star + \sqrt{\frac{2\tau^2}{n}\log\big(\frac{2}{q_f^\star(\delta, J_f(R_W))}\big)}\Big\} \le \delta.$$

The proof and specializations can be found in [1].[2]

For $\gamma \ge 1$, define $\mathsf{E}_\gamma(P\|Q) := \sup_A\{P(A) - \gamma Q(A)\}$ and let $\Delta_\gamma(R_W) := \mathsf{E}_\gamma(P_{WS}\|R_W P^n)$. Applying it to E_ϵ gives

$$P_{WS}(E_\epsilon) \le \gamma Q_R(E_\epsilon) + \Delta_\gamma(R_W) \le \gamma\theta_\epsilon + \Delta_\gamma(R_W).$$

Consequently, if $\Delta_\gamma(R_W) < \delta$, then

$$\Pr\Big\{|\,\mathrm{gen}(W,S)| > c^\star + \sqrt{\frac{2\tau^2}{n}\log\big(\frac{2\gamma}{\delta - \Delta_\gamma(R_W)}\big)}\Big\} \le \delta.$$

Remark 1. The same DPI step can also sharpen the Rényi-divergence bounds [15]. We omit this specialization due to space limitations, as the underlying argument is analogous.

[2]We treat the hockey-stick divergence separately as $f_\gamma(t) = (t-\gamma)_+$ is not strictly convex. This is a minor issue and does not affect our main results.

B. Sibson–Rényi Information and Maximal Leakage

We consider an alternative class of measures. For finite $\mathcal{W}$ and $1 < \beta < \infty$, define $I_\beta^{\mathrm{Sib}}(S;W) := \inf_{R_W} D_\beta(P_{WS}\|Q_R)$, where $D_\beta(P\|Q)$ denotes the Rényi-divergence, $\beta > 1$, $Q_R = R_W P^n$ and the infimum is over laws with $\mathrm{supp}(R_W) = \mathrm{supp}(P_W)$. Its infinite-order limit is the maximal leakage [29]:

$$I_\infty^{\mathrm{Sib}}(S;W) = \mathcal{L}(S \to W) := \log \sum\nolimits_{w\in\mathcal{W}} \operatorname*{esssup}_{s\sim P^n} P_{W|S}(w|s).$$

Let $d_\beta(u\|v) := \frac{1}{\beta-1}\log\big[u^\beta v^{1-\beta} + (1-u)^\beta (1-v)^{1-\beta}\big]$ for $v \in (0,1)$. For $\theta \in (0,1)$, $J \ge 0$ and $1 < \beta < \infty$, define $\mathfrak{R}_\beta(\theta, J) := \sup\{u \in [\theta,1] : d_\beta(u\|\theta) \le J\}$. For the trivial endpoint, set $\mathfrak{R}_\beta(1, J) := 1$. We have the following result.

Theorem 2. *For finite $\mathcal{W}$ and if the uniform reference-tail estimate* (5) *holds, for every $1 < \beta < \infty$ and $\epsilon > 0$,*

$$P_{WS}(E_\epsilon) \le \mathfrak{R}_\beta\big(\theta_\epsilon, I_\beta^{\mathrm{Sib}}(S;W)\big). \tag{8}$$

Consequently, for every $\delta \in (0,1)$, $J \ge 0$, let $q_\beta^\star(\delta, J) \in (0,\delta]$ be the unique solution of $d_\beta\big(\delta\|q_\beta^\star(\delta,J)\big) = J$, we have

$$\Pr\Big\{|\,\mathrm{gen}(W,S)| > c^\star + \sqrt{\frac{2\tau^2}{n}\log\big(\frac{2}{q_\beta^\star\big(\delta, I_\beta^{\mathrm{Sib}}(S;W)\big)}\big)}\Big\} \le \delta, \tag{9}$$

By $I_\infty^{\mathrm{Sib}}(S;W) = \mathcal{L}(S \to W)$, we have that, $P_{WS}(E_\epsilon) \le \min\{1, e^{\mathcal{L}(S\to W)}\theta_\epsilon\}$, and for each $\delta \in (0,1)$,

$$\Pr\Big\{|\,\mathrm{gen}(W,S)| > c^\star + \sqrt{\frac{2\tau^2}{n}\big(\mathcal{L}(S\to W) + \log\frac{2}{\delta}\big)}\Big\} \le \delta.$$

IV. Examples

We illustrate the improvement from the classical change-of-measure refinements of Section III using two quantum-learning examples with $c^\star > 0$ where the quantum states and loss observable let $c^\star$ be evaluated explicitly. All bounds use the same quantum correction and the reference-tail (5).

First consider their common quantum structure. Let $S = (Z_1, \ldots, Z_n) \sim P^n$ and $P_{W|S}$ be some learning algorithm. Let $b : \mathcal{Z} \to \{0,1\}$ encode a binary property of z, and

$$|\Phi_b\rangle := (I_2 \otimes X^b)\big((|00\rangle + |11\rangle)/\sqrt{2}\big), \quad \Phi_b := |\Phi_b\rangle\langle\Phi_b|.$$

For each i, let T_i and R_i denote the test and training qubits, respectively. Conditioned on $s = (z_1, \ldots, z_n)$, the quantum data consist of n test-training pairs, $\rho(s) = \bigotimes_{i=1}^n \Phi_{b(z_i)}^{T_i R_i}$. The learner uses the valid sample-dependent POVM $E_s(w) = P_{W|S}(w|s) I_{R^n}$ and retains R_i as a hypothesis qubit H_i. The induced classical channel is $P_{W|S}$, while the actual and decoupled states are $\sigma(w,s) = \bigotimes_{i=1}^n \Phi_{b(z_i)}^{T_i H_i}$ and $\omega(w,s) = (I_4/4)^{\otimes n}$. Write $\ell_0(w,z) \in [0,1]$ for its numerical task loss, distinguishing it from the quantum loss observable. For $\lambda \in (0,1)$, define the local quantum loss and its average by

$$\begin{aligned} L_\lambda(w,z) &= (1-\lambda)\ell_0(w,z) I_4 + \lambda(I_4 - \Phi_{b(z)}), \\ \widehat{L}_\lambda(w,s) &= (1/n)\sum\nolimits_{i=1}^n L_\lambda^{T_i H_i}(w, z_i), \end{aligned} \tag{10}$$

where each summand acts as the identity on the other pairs.

Since $\mathrm{Tr}[(I_4 - \Phi_b)\Phi_b] = 0$ and $\mathrm{Tr}[(I_4 - \Phi_b)I_4/4] = 3/4$, we let $\widehat{\ell}_0(w,s) = \frac{1}{n}\sum_{i=1}^n \ell_0(w,z_i)$ and $\ell_0(w) = \mathbf{E}_{Z\sim P}[\ell_0(w,Z)]$, and have

$$\widehat{\ell}_{\rho,\lambda}(w,s) = (1-\lambda)\widehat{\ell}_0(w,s), \ell_{\rho,\lambda}(w) = (1-\lambda)\ell_0(w) + \frac{3\lambda}{4},$$
$$\mathrm{gen}_\lambda(w,s) = (1-\lambda)\big(\ell_0(w) - \widehat{\ell}_0(w,s)\big) + 3\lambda/4. \tag{11}$$

The difference between the actual and decoupled quantum states gives a nonzero bias $3\lambda/4$. Both one-qubit marginals of Φ_b equal $I_2/2$. $\overline{D}_\alpha(\sigma(w,s)\|\omega(w,s)) = n\log 4$ for $0<\alpha<1$. The loss observable has local spectral width λ. The fixed-w empirical loss is an average of $[0,1-\lambda]$-valued r.v.. Hence

$$\mu_\lambda = \lambda/2, \quad c^\star = \lambda\sqrt{\log 2} > 0, \quad \tau_\lambda = (1-\lambda)/2. \tag{12}$$

Fixing $\lambda = 0.05, \delta = 0.05$ gives $c^\star = 0.041628, \tau_\lambda = 0.475$.

A. Bell-Augmented QNN Selection

We use the Breast Cancer Wisconsin Diagnostic dataset (569 labeled observations) [30]. We use 256 observations for a set of $K = 64$ fixed two-layer, single-qubit data-reuploading QNN classifiers, and P for the uniform distribution on the rest. An independent sample $S \sim P^{16}$ is used to select a classifier. For $z = (x,y)$, where $x \in \mathbb{R}^2$ is obtained by standardization and PCA fitted on the design split, and $y \in \{0,1\}$, let $p_w(1|x)$ be the predicted class-one probability of candidate w. We use the Brier task loss and Bell-state encoding

$$\ell_0(w,z) = \big(p_w(1|x) - y\big)^2, \qquad b(z) = y.$$

By uniform $\pi(w) = 1/K$, the learner is the Gibbs selector

$$P_{W|S}(w|s) = \frac{\pi(w)\exp[-\eta n\widehat{\ell}_{\rho,\lambda}(w,s)]}{\sum_{v\in\mathcal{W}}\pi(v)\exp[-\eta n\widehat{\ell}_{\rho,\lambda}(v,s)]},$$

where $\eta = 1.52737$ is the Gibbs inverse-temperature parameter. At confidence 95%, the estimated generalization radii are $r_{\mathrm{W4}} = 0.460103$ and $r_{\mathrm{W5}} = 0.424858$ for [15, Theorems 4,5], while Theorem 1 with hockey-stick gives $r_{\mathrm{HS}} = 0.406428$.

TABLE I
95% GENERALIZATION RADIUS. THEOREM 1 GIVES THE BEST RESULT.

Thm.4 of [15]	Thm.5 of [15]	Sibson–Rényi	Hockey-stick
0.46010	0.42486	0.40842	**0.40643**

B. Multi-Tier Randomized-Reveal Channel

Let $Z_i \in \{0,\dots,8\}$ be i.i.d. with $P(Z_i = 0) = 1-m$ and $P(Z_i = j) = ma_j$ for $1 \le j \le 8$, where $a_j = \frac{0.7^{j-1}}{\sum_{k=1}^8 0.7^{k-1}}$. For $s = (z_1,\dots,z_n)$, define a classifier $W \in \{0,\dots,8\}$ by

$$P_{W|S}(w|s) = \begin{cases} u_w, & z_1 = 0, \\ (1-\eta_{z_1})u_w + \eta_{z_1}\mathbf{1}\{w = z_1\}, & z_1 \ge 1. \end{cases}$$

where $u = (0.4, 0.075, \dots, 0.075)$ and $(\eta_1,\dots,\eta_8) = (0.20, 0.28, 0.36, 0.44, 0.70, 0.80, 0.90, 0.97)$. We take $n = 100$ and use $\ell_0(w,z) = \mathbf{1}\{w \ne z\}$, $b(z) = \mathbf{1}\{z \ne 0\}$ in (10).

We evaluate the bounds by enumerating this 9×9 joint law. At $m = 0.093$, [15, Theorems 4,5] give $r_{\mathrm{W4}} = 0.202362$ and $r_{\mathrm{W5}} = 0.197013$, whereas our hockey-stick bound gives $r_{\mathrm{HS}} = 0.188575$. On the evaluated grid, hockey-stick gives the smallest displayed radius for $m \in [0.002, 0.114]$; see Fig. 2.

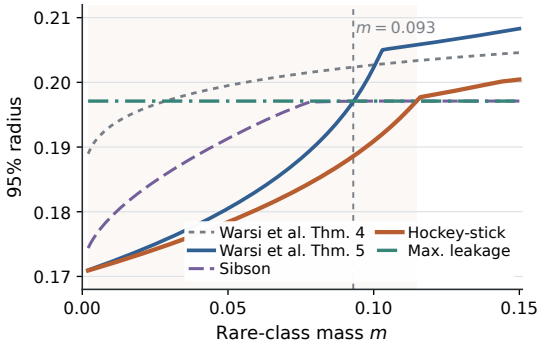


Fig. 2. Multi-tier randomized-reveal channel.

V. Quantum Reference-Tail Refinements

While Section III improves the classical layer, in this section we refine the quantum layer. Recall (4) gives $\mathrm{gen}(w,S) = b_w + m_w - \widehat{\ell}_\rho(w,S)$, and a sharper bound on the quantum bias b_w yields a sharper reference-tail. We focus on a loss-sensitive correction utilizing the quantum hockey-stick divergence [23].

A. Quantum Hockey-Stick Divergence

For density operators ω, σ and $\gamma \ge 1$, if $(\cdot)_+$ denotes the positive part of a Hermitian operator, we define the quantum hockey-stick divergence [23] by

$$\mathsf{H}_\gamma(\omega\|\sigma) := \mathrm{Tr}[(\omega - \gamma\sigma)_+] = \sup_{0\preceq M\preceq I}\mathrm{Tr}\,[M(\omega - \gamma\sigma)].$$

For all (w,s), we have loss observable

$$\widehat{L}(w,s) = a(w,s)I + d(w,s)M(w,s) \tag{13}$$

where $d(w,s) \ge 0$ and $0 \preceq M(w,s) \preceq I$. For a bounded Hermitian loss, this decomposition can be obtained by normalizing its spectral range. We then have the following result.

Proposition 3. *Under* (13)*, define*

$$B^+_{\mathrm{QHS}}(w) := \mathbf{E}_{S\sim P^n}\big[d(w,S)\inf_{\gamma\ge 1}\big\{\mathsf{H}_\gamma\left(\omega(w,S)\|\sigma(w,S)\right) + (\gamma-1)\,\mathrm{Tr}\,[M(w,S)\sigma(w,S)]\big\}\big].$$

$B^-_{\mathrm{QHS}}(w)$ is defined by switching $\omega(w,S)$ and $\sigma(w,S)$. We have

$$-B^-_{\mathrm{QHS}}(w) \le \ell_\rho(w) - m_w \le B^+_{\mathrm{QHS}}(w). \tag{14}$$

$B^+_{\mathrm{QHS}}(w)$ shifts the lower empirical-loss tail corresponding to $\mathrm{gen}(w,S) > \epsilon$; $B^-_{\mathrm{QHS}}(w)$ shifts the upper tail corresponding to $\mathrm{gen}(w,S) < -\epsilon$, proved in [1]. The correction is *loss-sensitive*: it depends on the state pair and the loss effect $M(w,s)$ via its acceptance probability under reference states.

B. Refined Reference-Tail

To get high-probability bound from (14), assume for fixed w, $\widehat{\ell}_\rho(w,S) = \frac{1}{n}\sum_{i=1}^n X_w(Z_i)$ for $Z_i \overset{\text{i.i.d.}}{\sim} P$. Set $m_w := \mathbf{E}[X_w(Z)]$ and, for $u \geq 0$, let $\psi_w^+(u) := \log \mathbf{E}e^{u(X_w(Z)-m_w)}$ and $\psi_w^-(u) := \log \mathbf{E}e^{u(m_w-X_w(Z))}$ with one-sided Chernoff rate functions $I_w^\pm(x) := \sup_{u\geq 0}\{ux - \psi_w^\pm(u)\}$. We have:

Theorem 4. *For $\epsilon > 0$, define*

$$\theta_w^{\text{ref}}(\epsilon) := \min\left\{1, e^{-nI_w^-\left(\epsilon - B_{\text{QHS}}^+(w)\right)} + e^{-nI_w^+\left(\epsilon - B_{\text{QHS}}^-(w)\right)}\right\}.$$

Then we have $P^n(E_{\epsilon,w}) \leq \theta_w^{\text{ref}}(\epsilon)$. For auxiliary R_W, define $\Theta_R(\epsilon) := \mathbf{E}_{W\sim R_W}\left[\theta_W^{\text{ref}}(\epsilon)\right]$, and then for every $\gamma \geq 1$,

$$P_{WS}(E_\epsilon) \leq \gamma\Theta_R(\epsilon) + \mathsf{E}_\gamma\left(P_{WS}\|P^n R_W\right). \tag{15}$$

The proof can be found in [1]. By finding a smaller ϵ for which the right-hand side of (15) is at most δ, we obtain a tighter generalization radius. The bounds from Section III apply with the uniform reference-tail θ_ϵ replaced by $\Theta_R(\epsilon)$.

If $X_w \in [0,1]$ and $B_{\text{QHS}}^\pm(w) \leq B(w)$, Hoeffding gives $P^n(E_{\epsilon,w}) \leq \min\left\{1, 2e^{-2n(\epsilon-B(w))^2}\right\}$ for $\epsilon > B(w)$; hence the quantum hockey-stick analysis can replace the uniform correction $c^\star$ by a loss- and hypothesis-dependent correction before the classical change-of-measure machinery.

VI. Examples Revisited

We revisit the two examples in Section IV to evaluate the additional improvement from Section V. The quantum states, losses, learning channels, and $c^\star, \lambda$ remain unchanged.

Recall the Bell-augmented loss in (10). For each pair, take $M_b = I_4 - \Phi_b$. Since $\text{Tr}[M_b\Phi_b] = 0$ and $\mathsf{H}_\gamma(I_4/4\|\Phi_b) = 3/4$ for every $\gamma \geq 1$, applying Proposition 3 locally and averaging over the n pairs gives the exact positive quantum bias

$$b_w = \ell_{\rho,\lambda}(w) - \mathbf{E}_{S\sim P^n}\widehat{\ell}_{\rho,\lambda}(w,S) = \frac{3\lambda}{4} =: B = 0.0375 < c^\star.$$

The local decomposition is important: applying the quantum hockey-stick inequality only to the full tensor-product state can be looser. For this construction the local infimum is already attained at $\gamma = 1$. Because the sign $b_w = B > 0$ is also known from (11), we can refine the two tails separately. For a fixed w, one-sided Hoeffding–Chernoff bounds give, for $\epsilon > B$,

$$P^n(E_{\epsilon,w}) \leq \theta_{\text{sH}}(\epsilon) := \min\left\{1, e^{-\frac{2n(\epsilon-B)^2}{(1-\lambda)^2}} + e^{-\frac{2n(\epsilon+B)^2}{(1-\lambda)^2}}\right\}. \tag{16}$$

This is a sign-aware case of the argument in Theorem 4.

A. Bell-Augmented QNN Selection

For the QNN example of Section IV-A, we keep the same Gibbs channel and $n = 16$. Rather than evaluating a model-specific Chernoff rate for each circuit, we use above sign-aware Hoeffding bound (16). It combines the known quantum bias B with the boundedness of the Brier loss under the i.i.d. reference sample, and therefore does not require the full per-circuit loss distribution. See Table II for the improvements.

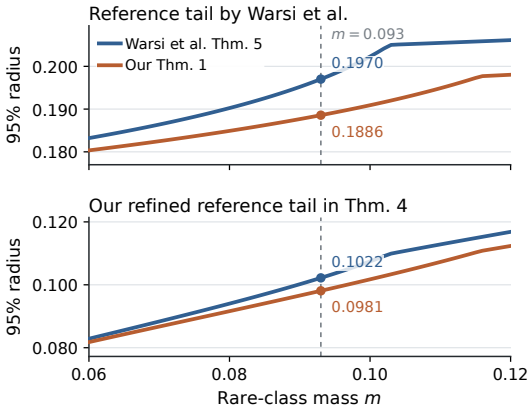


Fig. 3. Four-way comparison for the multi-tier channel as the rare-class mass m varies. We show the separate improvements obtained via our refined outer (classical) inequalities and inner (quantum) reference tails over [15, Thm. 5].

TABLE II
FOUR-WAY COMPARISON AT 95% GENERALIZATION RADII. WE SEPARATE THE IMPROVEMENTS FROM OUR CLASSICAL LAYER (THEOREM 1) AND QUANTUM LAYER (THEOREM 4) ON THE TWO EXAMPLES.

	[15, Thm. 5]	Thm. 1	Thm. 4	Combined
QNN	0.42486	0.40643	0.39876	**0.37962 (-10.6%)**
Multitier	0.19701	0.18858	0.10220	**0.09810 (-50.2%)**

B. Multi-Tier Randomized-Reveal Channel

For the multi-tier channel in Section IV-B, the fixed-w loss distribution is known. Let $p_w = P(Z = w)$, $N_w \sim \text{Bin}(n, p_w)$ under the product reference law. By (11), $\text{gen}_\lambda(w,S) = B + (1-\lambda)(N_w/n - p_w)$. The Bernoulli Chernoff rate function gives

$$\theta_w^{\text{Ber}}(\epsilon) := \min\left\{1, e^{-nd(p_w + \frac{\epsilon-B}{1-\lambda}\|p_w)} + e^{-nd(p_w - \frac{\epsilon+B}{1-\lambda}\|p_w)}\right\}$$

for $\epsilon > B$, where $d(\cdot\|\cdot)$ is the binary relative entropy. A Chernoff term is set to zero when its threshold lies outside $[0,1]$. We substitute $\Theta_{P_W}(\epsilon) = \sum_w P_W(w)\theta_w^{\text{Ber}}(\epsilon)$ into the smooth-max event transfer [15] or the hockey-stick inequality (15).

We use Figure 3 and Table II to explicitly show the four-way comparison of the improvements achieved by our two refinements.

VII. Concluding Remarks

In this paper, we propose high-probability generalization bounds for a quantum learning framework. We refine its classical and quantum layers via an indicator-DPI trick and a loss-sensitive quantum hockey-stick inequality, respectively, and use two explicit examples to demonstrate our improvements.

We note that a related quantum learning framework has recently been proposed [17] through a single global classical–quantum divergence, without introducing a separate correction term such as $c^\star$. Understanding more precisely the relation between this global formulation and the two-layer decomposition considered in our paper (see also [14], [15]) is an interesting direction for future work. Another direction is to study more systematically the connection between the generalization bounds and quantum differential privacy [17].

VIII. Disclosure of Use of AI Tools

The authors used OpenAI Codex and GPT-5.6 Sol to assist with exploring proof strategies (Section V), numerical calculations, plotting computational figures (in Section IV and Section VI) and polishing the language. The authors verified the mathematical statements, proofs, numerical results, and figures in the final manuscript independently and take full responsibility for its contents.

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